PhysioCare

Digital Physiotherapy Management System

Comprehensive Project Guide & Complete Workflow Architecture

What is PhysioCare? (In Simple Words)

PhysioCare is an online healthcare website that connects patients who need physical therapy with certified physiotherapists (doctors).

Instead of waiting in long clinic lines, patients can book home or clinic visits, follow customized exercise recovery programs, watch guided therapy videos, download official clinical session reports, and chat with an intelligent AI assistant that answers physiotherapy questions 24/7.

1. Key Project Highlights

Everything that makes PhysioCare powerful and unique

- **Two Specialized Portals:** Separate dashboards for Patients and Doctors, each with dedicated permissions.
- **Smart OTP Authentication:** Real email OTP verification via EmailJS with a 60-second cooldown timer.
- **PhysioCare AI Assistant:** Instant recovery tips and exercise advice powered by Google Gemini streaming AI.
- **Clinical PDF Session Reports:** Official A4 clinical reports with digital clinic stamps, medicines, and physician signatures.
- **Exercise Video Therapy:** Rehabilitation video libraries where patients can log their pain level (1-10) after each session.
- **Rehabilitation Program Codes:** Doctors create multi-week therapy tracks with unique 6-character access codes for patients.

2. Technology Stack

Modern technologies used to build this platform

Frontend (User Interface)

- React 19 & Vite 7 (Super fast loading)
- Tailwind CSS & Custom Calm Theme
- React Router 7 (Smooth navigation)
- Responsive on Mobile, Tablet & Desktop
- Real-time BroadcastChannel sync

Backend, Database & APIs

- Node.js & Express 5 REST API
- MongoDB Database (Mongoose 9)
- Google Gemini (Streaming AI)
- EmailJS REST API (Live Gmail OTP)
- JWT Security & bcrypt password hash

3. How It Works: Step-by-Step User Journey

Clear guide on how Patients and Doctors use PhysioCare

Step 1: Account Creation & Secure Login

1. The user goes to the website and clicks 'Sign Up' as either Patient or Physiotherapist.
2. They enter their name, email address, and password.
3. PhysioCare immediately sends a real 6-digit One-Time Password (OTP) to their Gmail inbox.
4. The user enters the OTP within 10 minutes to verify their email address.
5. Once verified, a secure JWT session token is created, keeping their account logged in.

Step 2: Patient Experience (Recovery Made Simple)

- Find Doctors: Patients browse certified doctors in their city with experience & fees listed.
- Book Appointment: Pick a convenient date and time. Double-booking is automatically prevented.
- AI Recovery Chat: Open the floating PhysioCare AI bot at any time to ask for gentle stretches, exercise recommendations, posture guidance, and recovery tips in clean plain text.
- Therapy Regimens: Join specialized rehabilitation programs using a doctor's 6-letter code.
- Video Therapy: Watch guided exercise demonstrations and record personal pain levels (1 to 10).
- Medical Reports: Read and download clinical session reports issued by their treating doctor.

Step 3: Doctor Experience (Clinical Management)

- Clinical Dashboard: View upcoming patient bookings, confirm slots, or reschedule if needed.
- Issue Session Reports: After treating a patient, the doctor fills in clinical notes, prescribed medicines, exercises, suggestions, and a next follow-up date, then sends it to the patient.
- Program Studio: Create multi-week recovery programs with auto-generated patient invite codes.
- Video Demonstrations: Upload exercise videos (MP4, WebM, MOV) or clinical YouTube links.
- Health Articles: Write educational articles to teach patients healthy posture and habits.
- Private Case Notes: Maintain confidential clinical notes for each individual patient.

4. PhysioCare AI Recovery Assistant

24/7 Smart Clinical Guidance Powered by Google Gemini

- **Real-Time Streaming:** Answers stream word-by-word instantly using Server-Sent Events (SSE). No long waiting!
- **Strict Physiotherapy Scope:** The AI ONLY answers questions about physical therapy, stretches, rehabilitation, mobility, posture, and ergonomics. If asked about unrelated things (like coding, math, or politics), it politely declines and guides the user back to recovery.
- **Clean Plain-Text Formatting:** No confusing markdown symbols like asterisks (*) or hashtags (#). Clean numbered lists and simple headings.
- **Emergency Triage Guardrail:** If a patient mentions dangerous symptoms like chest pain, trouble breathing, or sudden numbness, the AI immediately instructs them to stop and call emergency services (112 or local hospital).

5. Clinical Medical Reports & PDF Architecture

Official Documentation Ready for Patient Files

- **A4 Letterhead Standard:** Designed with clinical branding, clinic stamp, issue date, and doctor identification.
- **Prescription & Directives:** Clearly displays clinical observations, prescribed medications, and recommended home exercises.
- **Follow-Up Scheduler:** Highlights the exact date for the next physical therapy check-up.
- **Print-Ready PDF Export:** Patients can click 'Print / Save as PDF' in the Reports tab to generate an exact medical record.

6. Security & Data Protection

Keeping Patient Records Safe and Private

- **Password Protection:** Passwords are encrypted using bcrypt hashing before being saved to the database.
- **Role-Based Access Control:** Patients cannot access doctor panels; doctors cannot modify other doctors' patient reports.
- **Ownership Validation:** Only treating doctors and the patient themselves can access private session reports and medical logs.
- **Rate Limiting:** Authentication endpoints and AI chat are protected against spam with automated rate limiters.

7. How to Run This Project Locally

Simple step-by-step commands for anyone testing the code

1. Clone the Repository

git clone https://github.com/rishavrjpt/PhysioCare.git
cd PhysioCare

2. Install All Dependencies

npm run install:all (Installs root, backend, and frontend packages)

3. Configure Environment Variables

Copy backend/.env.example to backend/.env and set your MongoDB URI and JWT Secret.

4. Start Development Servers

npm run dev (Runs both backend on port 8080 and frontend on port 5173 together)

5. Open in Browser

Visit http://localhost:5173 to test patient and doctor features!

8. Database Models Summary

How data is organized inside MongoDB

- **User:** Stores account details, role (patient/doctor), encrypted password, email verification OTP.
- **DoctorProfile:** Specialization, years of experience, clinic/hospital, consultation fee, available schedule.
- **Appointment:** Patient ID, Doctor ID, booking date & time, status (pending/confirmed/completed), review.
- **MedicalReport:** Doctor's clinical findings, prescribed medicine, home exercises, follow-up date.
- **Program:** Rehabilitation program title, description, specialty, and unique 6-character access code.
- **Video:** Exercise video title, description, file URL, duration, and associated program.
- **Article:** Educational blog post with title, category, cover image, and doctor author ID.

9. Summary & Contact

Project credits and repository link

Project Information & GitHub Repository

- Project Name: PhysioCare
- Lead Developer & Maintainer: Rishav Raj
- GitHub Profile: https://github.com/rishavrjpt
- GitHub Repository: https://github.com/rishavrjpt/PhysioCare
- Contact Email: rajrishav7013@gmail.com
- License: MIT Open Source License (2026)